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4-25-22

*** There does not seem to be a variable to restrict the data to three-person households

*** I had no idea how to do most of the stata coding for this homework

1.

Code/Output:

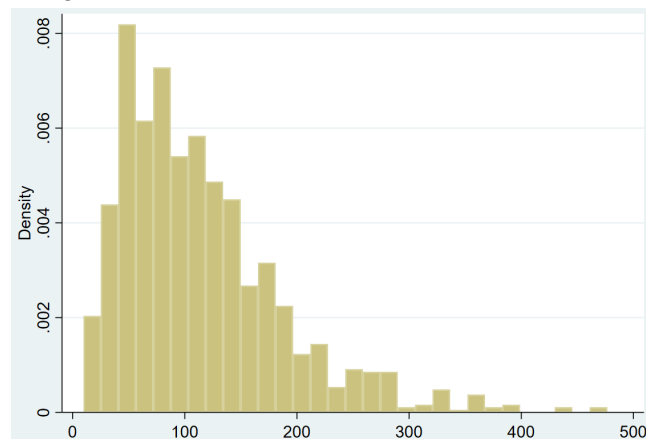
```
summarize food
```

Variable	Obs	Mean	Std. dev.	Min	Max
food	1,200	114.4431	72.6575	9.63	476.67

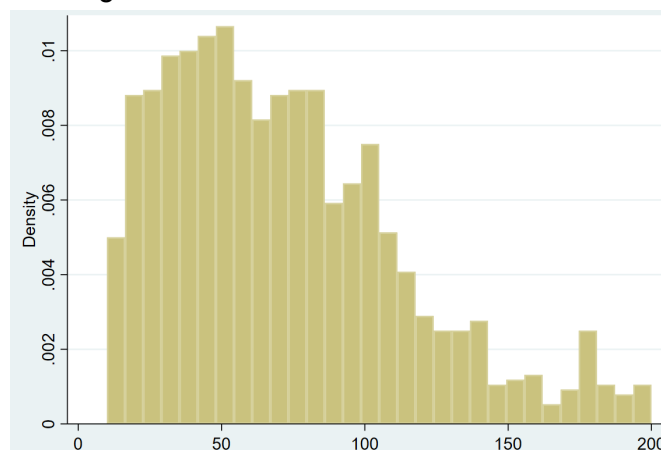
```
summarize income
```

Variable	Obs	Mean	Std. dev.	Min	Max
income	1,200	72.14264	41.65228	10	200

Food histogram code: histogram food



Income histogram code: histogram income



Neither histogram is symmetrical, both appear to have the majority of their data below their respective means. Both sample means are larger than the median ($114.5 > 99.8$, $72.14 > 65.29$, respectively), meaning the distributions are skewed

```
. sktest income
```

Skewness and kurtosis tests for normality

Variable	Obs	Pr(skewness)	Pr(kurtosis)	Joint test	
				Adj chi2(2)	Prob>chi2
income	1,200	0.0000	0.0358	92.39	0.0000

2.

Code/outputs:

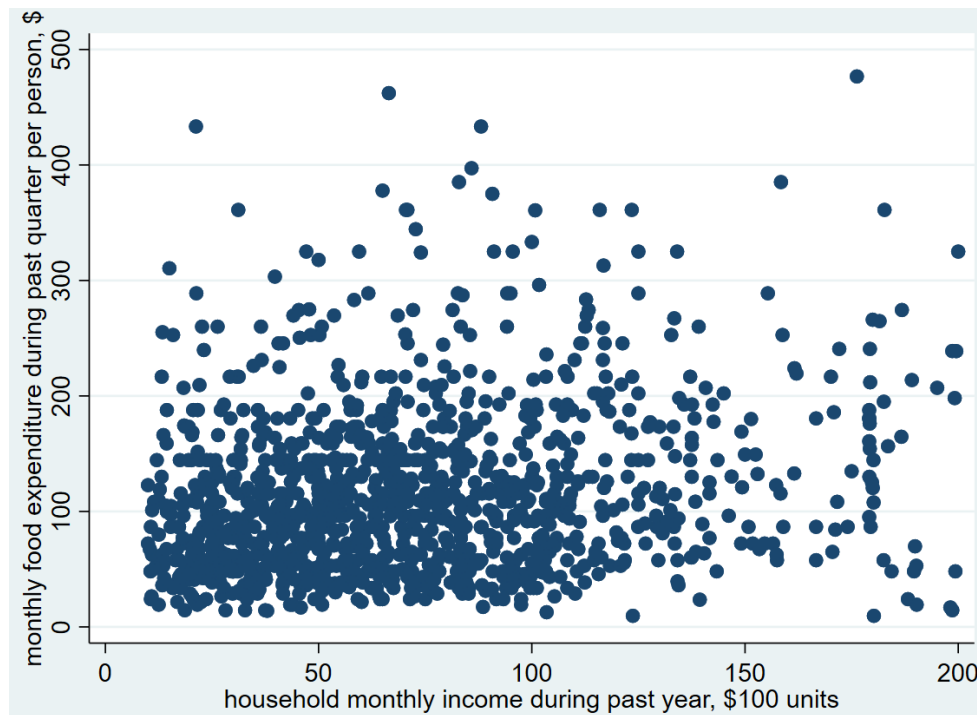
```
. regress food income
```

Source	SS	df	MS	Number of obs	=	1,200
Model	267625.41	1	267625.41	F(1, 1198)	=	52.89
Residual	6062029.78	1,198	5060.12503	Prob > F	=	0.0000
				R-squared	=	0.0423
				Adj R-squared	=	0.0415
Total	6329655.19	1,199	5279.11192	Root MSE	=	71.135

food	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
income	.3586867	.049321	7.27	0.000	.2619215	.455452
_cons	88.5665	4.108188	21.56	0.000	80.50646	96.62654

95% confidence interval for B2: (.2619215, .455452)

Code: scatter food income

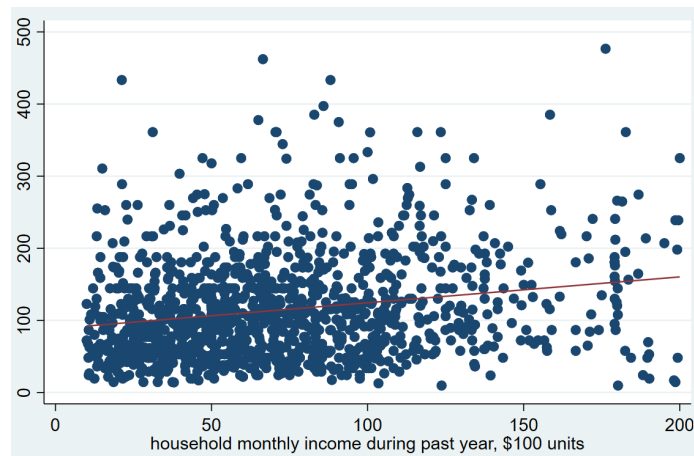


This does appear to be relatively precise since the confidence interval is relatively narrow.

3.

Code: twoway scatter food income || lfit food income

Output:



It would be more important that the error is normally distributed. If the error deviates greatly from the mean, then it becomes more difficult to conclude the significance of household income in predicting food expenditure.

Code/Output:

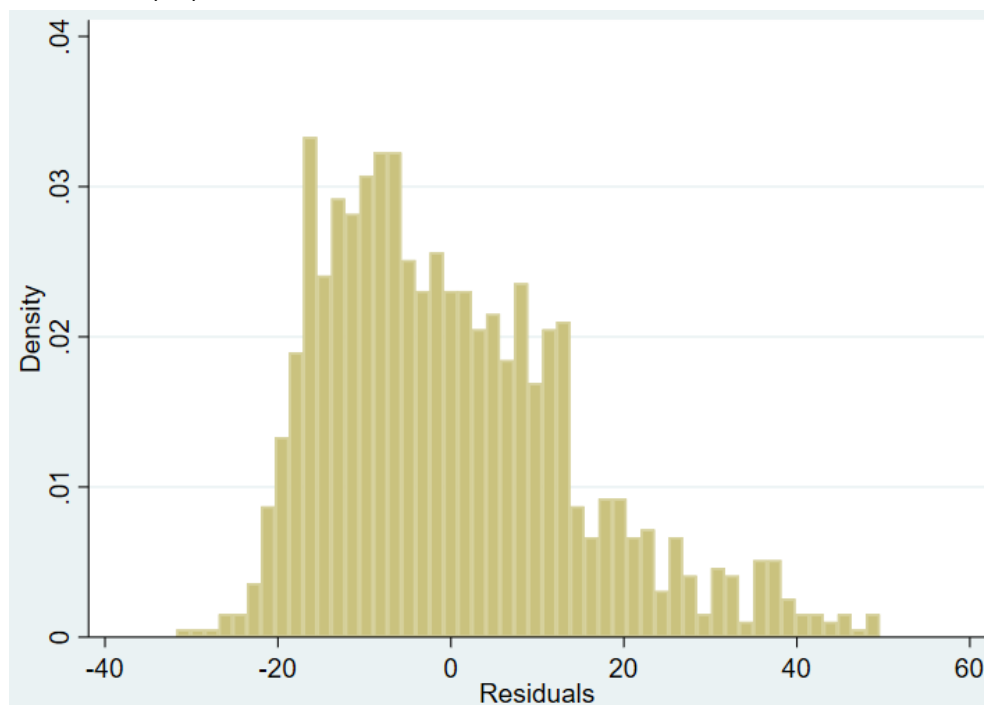
```
gen y1 = 88.5665+0.3586867*income
```

```
reg y1 food
```

```
predict y1hat
```

```
predict error1, resid
```

```
histogram error1, bin(50)
```



```
display "JB TS= "(_N/6)*(r(skewness)^2+(((r(kurtosis)-3)^2)/4))
```

4. elasticity CIs

Code/Output:

```
summarize income
```

Variable	Obs	Mean	Std. dev.	Min	Max
income	1,200	72.14264	41.65228	10	200

```
local meaninc1 = r(mean)
```

```
summarize food
```

Variable	Obs	Mean	Std. dev.	Min	Max
food	1,200	114.4431	72.6575	9.63	476.67

```
local meanfoo1 = r(mean)
```

```
local f1 = `meaninc1'*_b[income]+`meanfoo1'*_b[food]+_b[_cons]
```

```
display "income: eyex = "(`meaninc1*_b[income])/`f1'
```

```
display "food: eyex = "(`meanfoo1*_b[food])/`f1'
```

```
margins, eyex(income food) atmeans nose
```

```
margins, eyex(income) at (income = (19) (65) (160))
```

```
* this code does not seem to be working
```

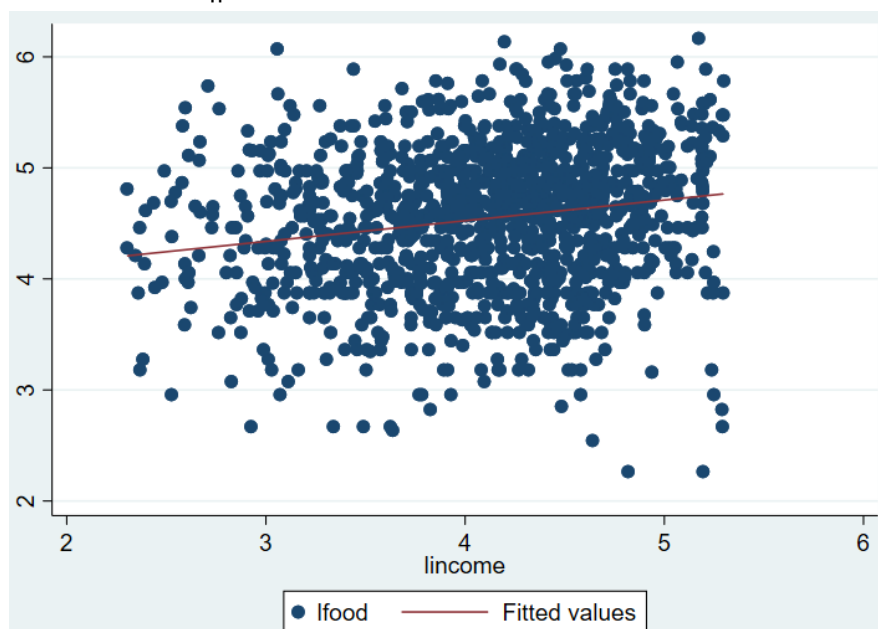
5.

code/output:

```
gen lfood = ln(food)
```

```
gen lincome = ln(income)
```

```
twoway scatter lfood lincome || lfit lfood lincome
```



reg lfood lincome

Source	SS	df	MS	Number of obs	=	1,200
Model	16.9590002	1	16.9590002	F(1, 1198)	=	41.18
Residual	493.406139	1,198	.411858213	Prob > F	=	0.0000
				R-squared	=	0.0332
				Adj R-squared	=	0.0324
Total	510.365139	1,199	.425658999	Root MSE	=	.64176

lfood	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
lincome	.1863054	.0290335	6.42	0.000	.1293432	.2432675
_cons	3.778932	.1203492	31.40	0.000	3.542814	4.015051

6. elasticity CIs

code/output:

summarize lincome

local meaninc2 = r(mean)

summarize lfood

local meanfoo2 = r(mean)

local f2 = `meaninc2' \* \_b[income] + `meanfoo2' * _b[food] + _b[_cons]

display "income: eyex = "(`meaninc2' * _b[income]) / f2'

display "food: eyex = "(`meanfoo2' * _b[food]) / f2'

margins, eyex(lincome lfood) atmeans nose

margins, eyex(lincome) at (lincome = (19) (65) (160))

7. (residual histogram)

code/output:

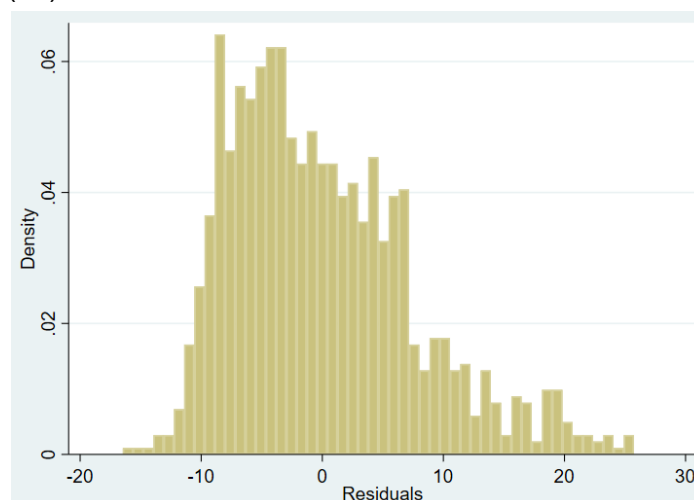
gen y2 = 3.77892 + 0.1862054 * income

reg y2 food

predict y2hat

predict error2, resid

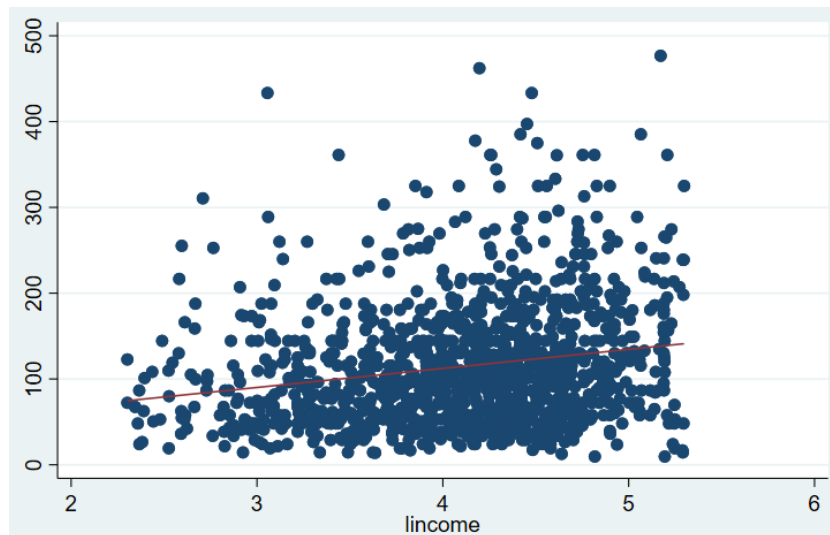
histogram error2, bin(50)



8.

code/output:

twoway scatter food lincome || lfit food lincome



9. Elasticity CIs

code/output: **sample means of log(income) and food already calculated in previous questions

local f3 = `meaninc2' \*\_b[income]+`meanfoo1'*_b[food]+_b[_cons]

display "income: eyex = "(`meaninc2\*\_b[income])/`f2'

display "food: eyex = "(`meanfoo1\*\_b[food])/`f2'

margins, eyex(lincome food) atmeans nose

margins, eyex(lincome) at (lincome = (19) (65) (160))

10. residuals plot

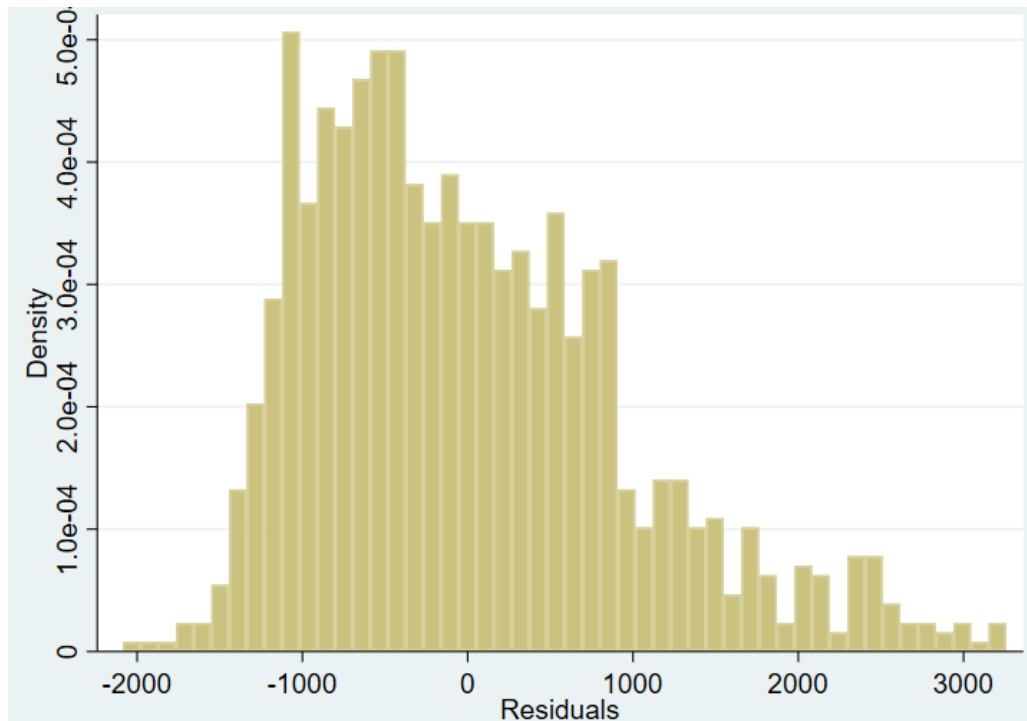
code/output:

reg food logincome

Source	SS	df	MS	Number of obs	=	1,200
Model	240525.871	1	240525.871	F(1, 1198)	=	47.32
Residual	6089129.32	1,198	5082.74568	Prob > F	=	0.0000
				R-squared	=	0.0380
				Adj R-squared	=	0.0372
Total	6329655.19	1,199	5279.11192	Root MSE	=	71.293

food	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
logincome	22.18738	3.225332	6.88	0.000	15.85946	28.51531
_cons	23.56848	13.3696	1.76	0.078	-2.661956	49.79892

```
gen y3 = 22.18738+23.56848*income  
reg y3 food  
predict y3hat  
predict error3, resid  
histogram error3, bin(50)
```



Extremely small densities? I probably did this all wrong